

Pod Name	Node	GPU	Startup (sec)
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-v95js	pokprod-b93r39s2	NVIDIA-H100-80GB-HBM3	148.0s
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-gccwx	pokprod-b93r43s2	NVIDIA-H100-80GB-HBM3	180.0s

EPP Configuration (Feature Gates & Scorer Weights)

apiVersion: inference.networking.x-k8s.io/v1alpha1
kind: EndpointPickerConfig
featureGates:
- flowControl
plugins:
- type: queue-scorer
- type: kv-cache-utilization-scorer
- type: prefix-cache-scorer
schedulingProfiles:
- name: default
 plugins:
 - pluginRef: queue-scorer
 weight: 2
 - pluginRef: kv-cache-utilization-scorer
 weight: 2
 - pluginRef: prefix-cache-scorer
 weight: 3

Benchmark Load Generator Configuration

Profile: chatbot_synthetic.yaml
target: http://infra-llmdbench-inference-gateway-istio.wva-bench-test.svc.cluster.local:80
model: Qwen/Qwen3-0.6B
request_type: text_completions
profile: constant
rate: [1,2,4,8]
max_seconds: 600
random_seed: 42
data:
 prompt_tokens_min: 10
 prompt_tokens_max: 8192
 prompt_tokens: 4096
 prompt_tokens_stdev: 2048
 output_tokens_min: 10
 output_tokens_max: 2048
 output_tokens: 1024
 output_tokens_stdev: 512
 samples: 1000

Autoscaling Configuration (HPA & VA)

Variant (VA) [workload-variant-autoscaler-va]: Min Replicas = 1, Max Replicas = 10, Cost Factor = 10.0
HorizontalPodAutoscaler (HPA) [workload-variant-autoscaler-hpa]: Min Replicas = 1, Max Replicas = 10
 > ScaleUp: Stabilization Window = 240s | Policies = [{"periodSeconds": 15, "type": "Percent", "value": 100}]
 > ScaleDown: Stabilization Window = 240s | Policies = [{"periodSeconds": 15, "type": "Percent", "value": 100}]

True Serving Capacity Analysis (GuideLLM results.json)

Rate: 1.0 RPS | Achieved: 0.99 RPS | Errors: 0.00 RPS | Tokens/s: 5143.30
Rate: 2.0 RPS | Achieved: 1.98 RPS | Errors: 0.00 RPS | Tokens/s: 10133.63
Rate: 4.0 RPS | Achieved: 3.82 RPS | Errors: 0.00 RPS | Tokens/s: 19740.57
Rate: 8.0 RPS | Achieved: 4.07 RPS | Errors: 1.86 RPS | Tokens/s: 21049.92

=> Estimated True Serving Capacity: ~3.82 RPS (Highest successful rate with 0 errors)

Understanding GuideLLM Metrics:
• Target Rate (RPS): The configured constant request rate the load generator attempts to send.
• Achieved RPS: The actual rate of requests that completed successfully with a full response.
• Error RPS: The rate of requests that failed (e.g., HTTP 5xx) or were strictly dropped/incomplete.
• True Serving Capacity: Evaluated as the highest Achieved RPS recorded just before the system reaches hardware saturation and begins generating Error RPS (dropped requests).

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Autoscaling Run Score (Lower is Better)

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Worst-Case P99 TTFT: 71109.53 ms
Worst-Case P99 ITL: 32.14 ms
Average Replicas: 1.05

Target SLAs: TTFT = 50.0ms | ITL = 50.0ms

Formula Engine:
1. TTFT Penalty = (Actual P99 TTFT) / (SLA TTFT)
2. ITL Penalty = (Actual P99 ITL) / (SLA ITL)
3. Cost Factor = VA Custom Resource declared Cost (10.0)
4. Avg Replicas Factor = Average running decode pods (1.05)
Autoscaling Score = Avg_Replicas × Cost_Factor × (TTFT_Penalty + ITL_Penalty)

Latency Penalty Subtotal = (71109.53/50.0) + (32.14/50.0) = 1422.83
Resource Multiplier = 1.05 × 10.0 = 10.53
=> Final Score = 10.53 × 1422.83 = 14981.60

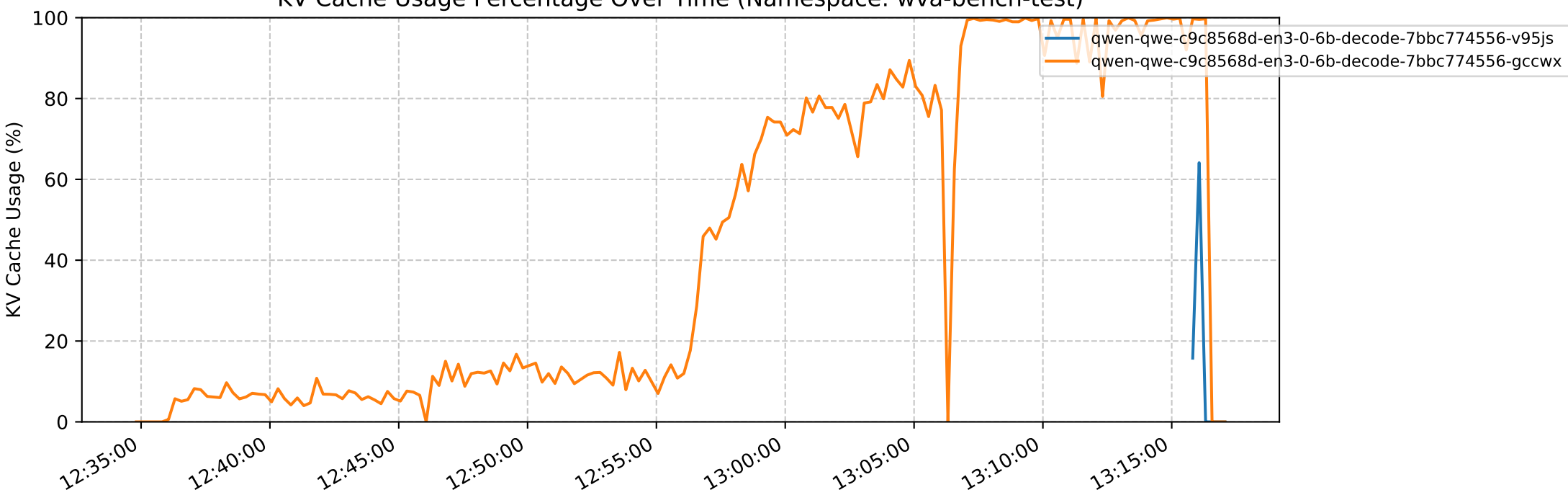
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GuideLLM Benchmark Inter-Phase Delay Analysis

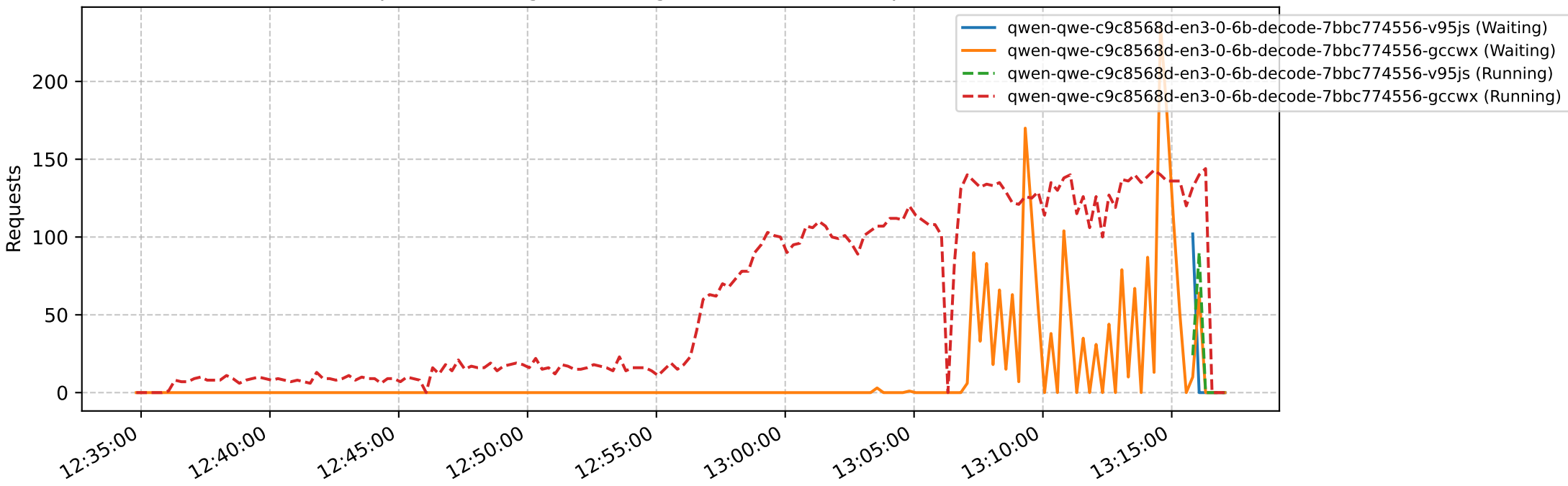
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Phase 0 (benchmark_report_results.json_0.yaml): Start = 1773776148.87, Stop = 1773776748.87 (Duration: 600.0s)
Gap between Phase 0 and 1: 3.64 seconds
Phase 1 (benchmark_report_results.json_1.yaml): Start = 1773776752.51, Stop = 1773777352.51 (Duration: 600.0s)
Gap between Phase 1 and 2: 5.73 seconds
Phase 2 (benchmark_report_results.json_2.yaml): Start = 1773777358.24, Stop = 1773777958.24 (Duration: 600.0s)
Gap between Phase 2 and 3: 9.35 seconds
Phase 3 (benchmark_report_results.json_3.yaml): Start = 1773777967.59, Stop = 1773778567.59 (Duration: 600.0s)

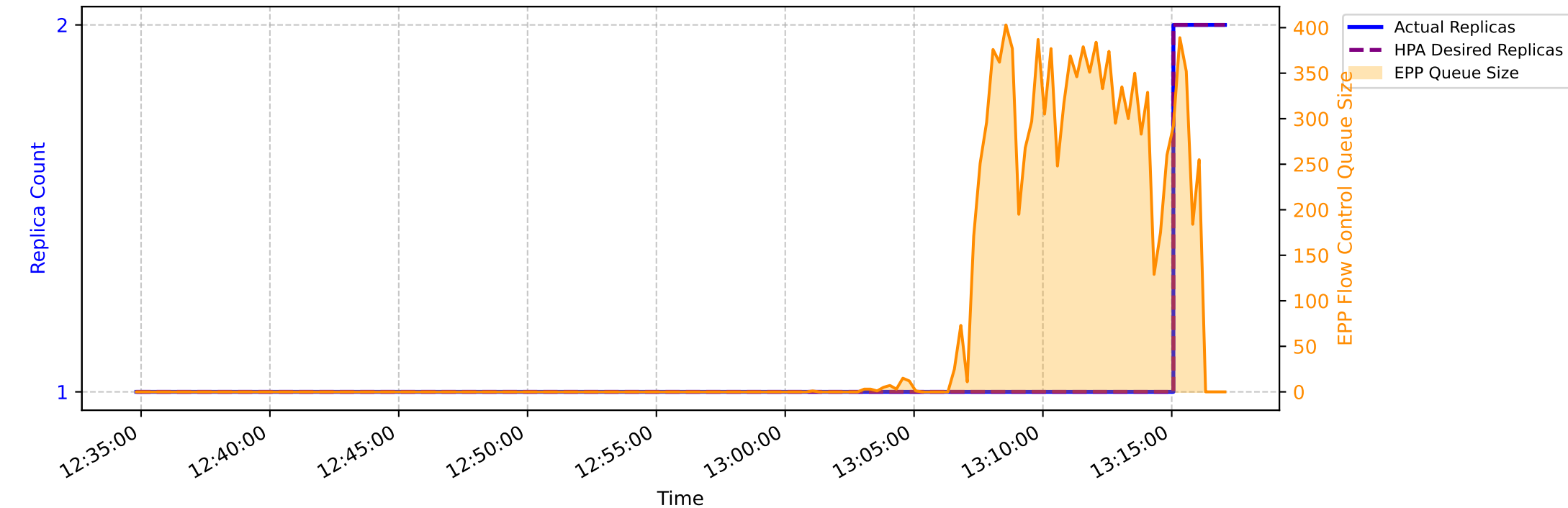
KV Cache Usage Percentage Over Time (Namespace: wva-bench-test)



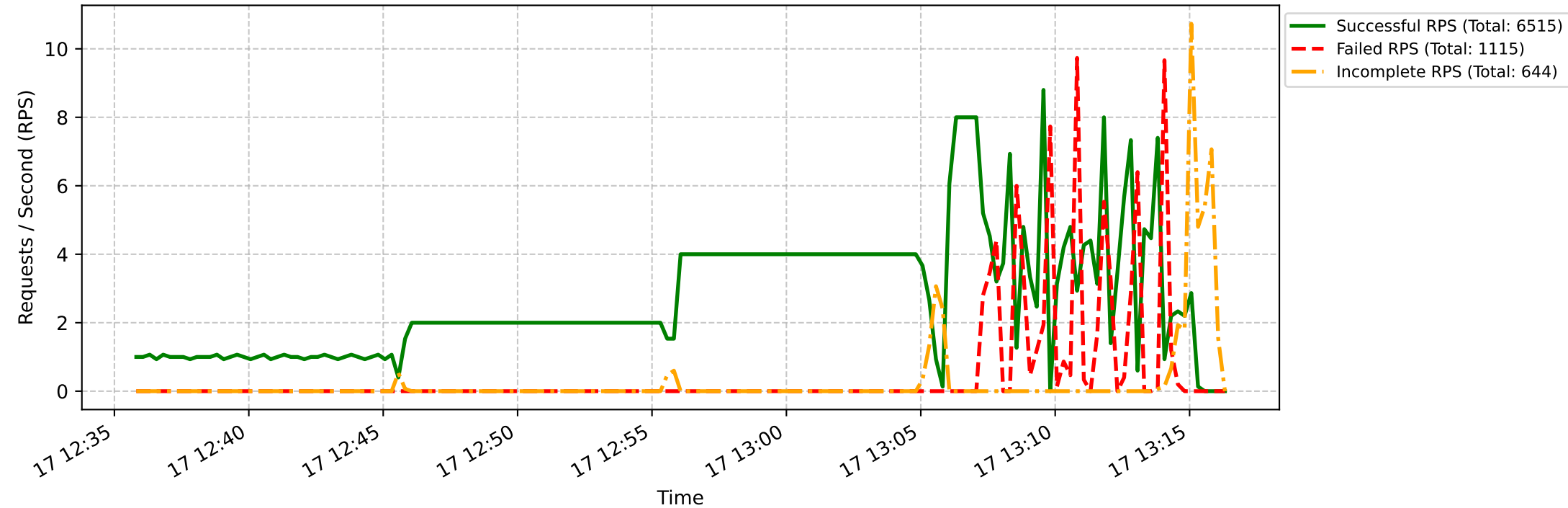
Number of Requests Waiting & Running Over Time (Namespace: wva-bench-test)



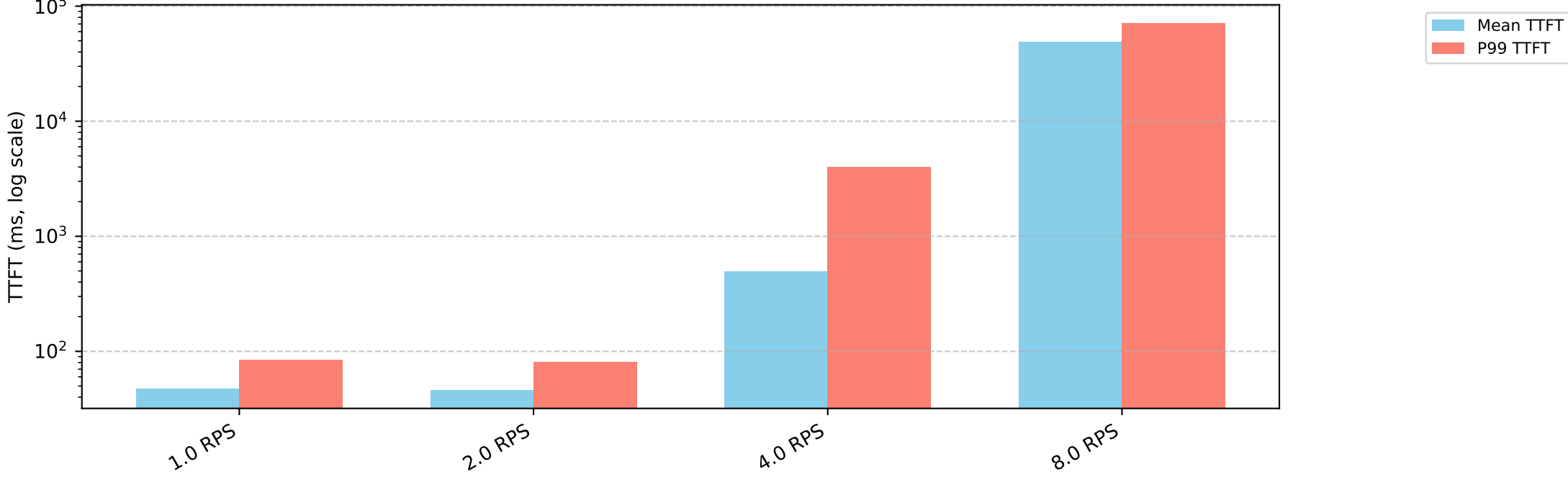
Decode Replica Count & EPP Queue Over Time (Namespace: wva-bench-test)



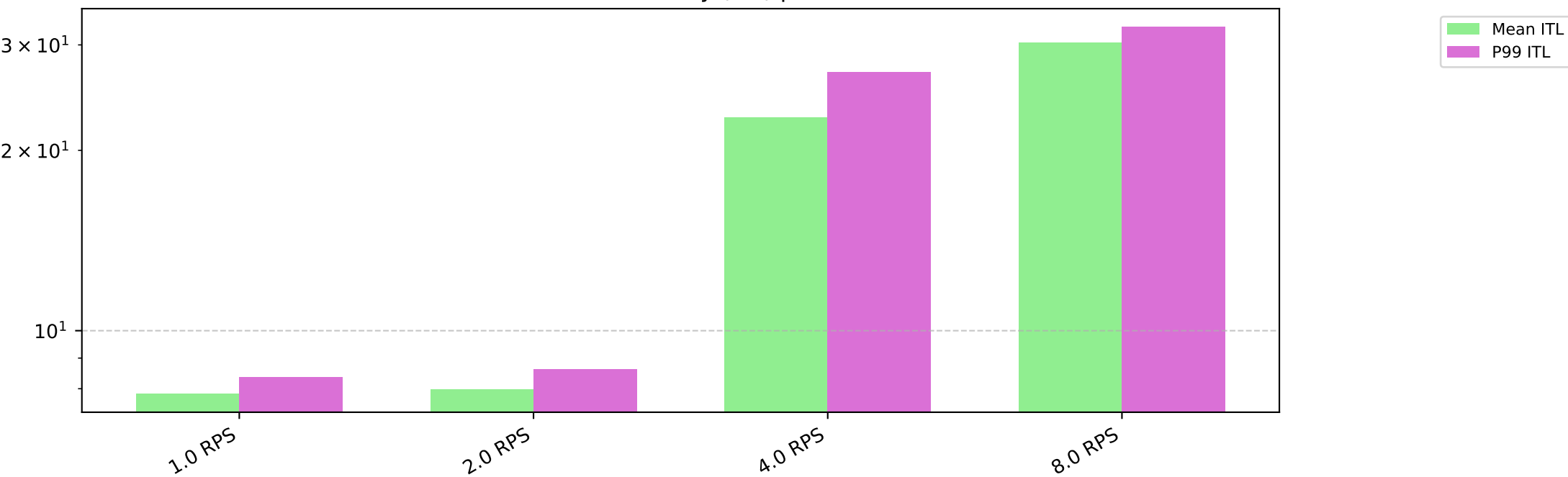
GuideLLM Requests (Succeeded vs Failed vs Incomplete) Over Time



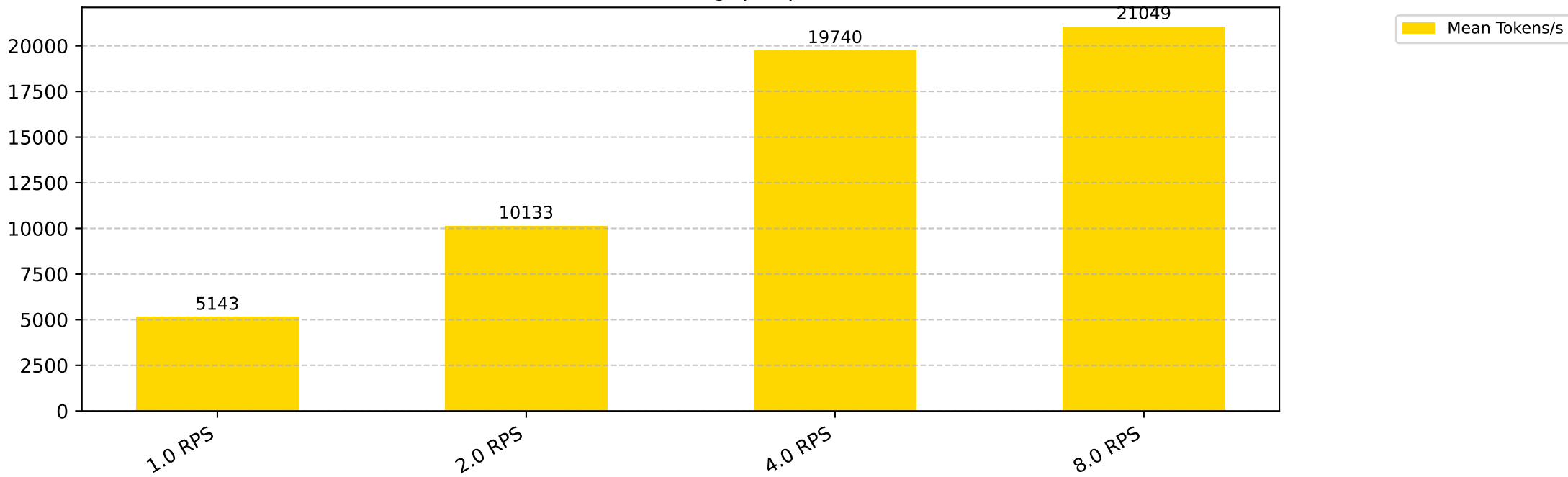
Time To First Token (TTFT) per Run



Inter-Token Latency (ITL) per Run



Overall Token Throughput per Run



Request Concurrency vs End-to-End Latency Profile

